

Assignment - 1

Program - 1 - Design and implementation of a FULL SUBTRACTOR using VHDL

⇒ Inputs : A, B, Bin

⇒ Outputs : Diff (Difference), Bout (Borrow out)

⇒ Truth table :

A	B	Bin	Diff	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

⇒ Program :

Library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity full_subtractor is

Port (

A : in STD_LOGIC; -- Input A

B : in STD_LOGIC; -- Input B

Bin : in STD_LOGIC; -- Borrow input

Diff : out STD_LOGIC; -- Difference output

Bout : out STD_LOGIC -- Borrow output

);

end full_subtractor;

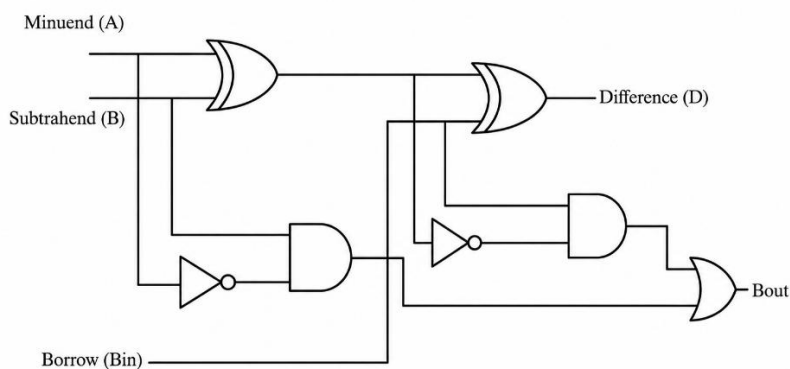
architecture Behavioral of full_subtractor is

begin

Diff <= A XOR B XOR Bin; -- Diff = $A \oplus B \oplus Bin$

Bout <= ((NOT A) AND B) OR -- Bout = $(\neg A \cdot B) + (Bin \cdot (A \text{ XNOR } B))$
 (Bin AND (A XNOR B));

end Behavioral;

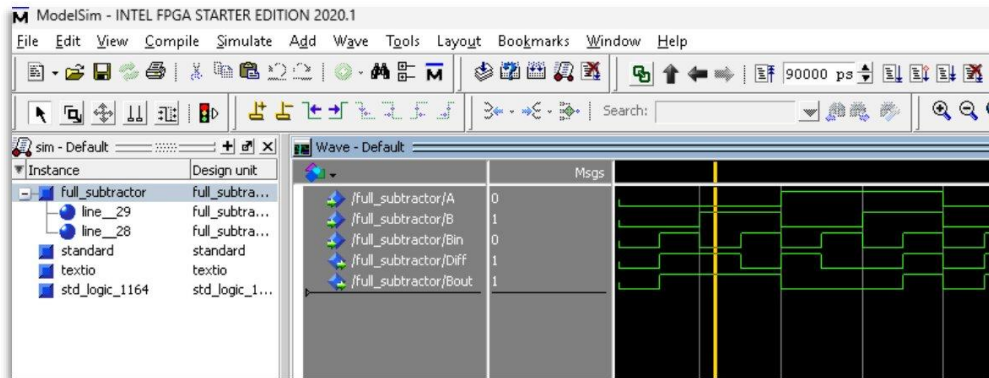


Full Subtractor

1. Simulation Using Clock Signals

The full subtractor was simulated using clock signals with a 50% duty cycle.

- A – 80 ns
- B – 40 ns
- Bin – 20 ns



2. Simulation Using Force Commands

The full subtractor was tested using force commands for all input combinations.

